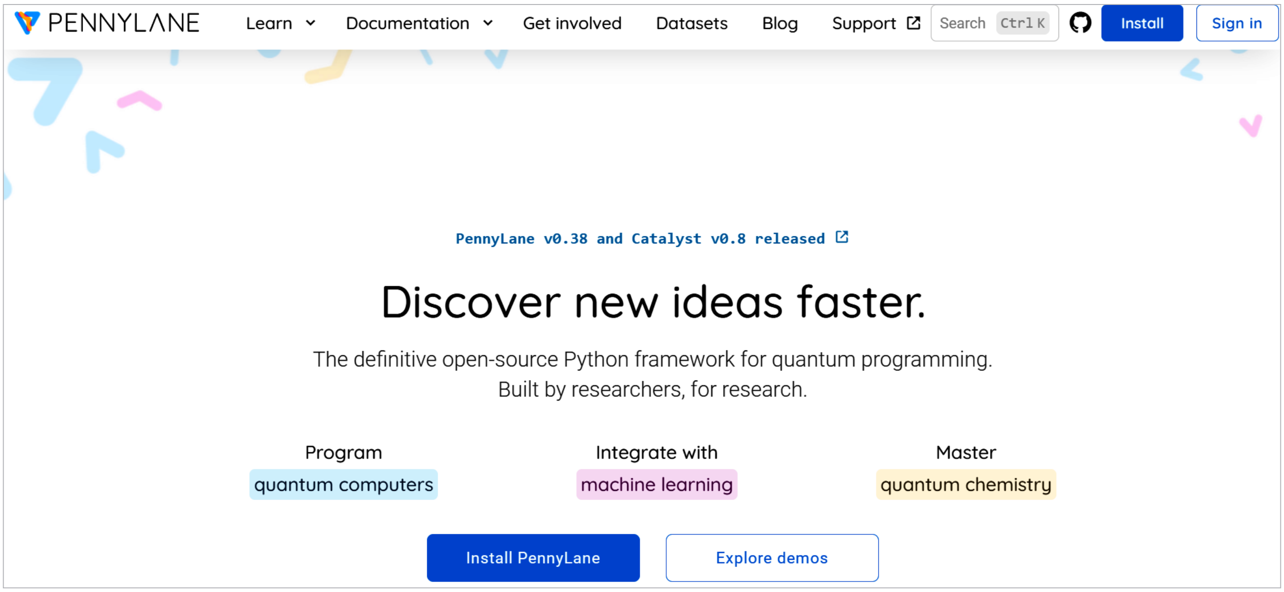

Figure 2: Pennylane

It can be programmed with assorted libraries and cloud platforms for executing quantum circuits, including IBM Qiskit, Google Cirq, Rigetti and Xanadu Strawberry Fields.

TensorFlow Quantum

https://www.tensorflow.org/quantum
Developed by Google, TensorFlow Quantum is an expansion of the TensorFlow library that works with the making of quantum AI models. It allows the flawless implementation of quantum computations and models for different applications in real world domains.

Qiskit

https://www.ibm.com/quantum/qiskit
Qiskit is an open source quantum processing platform developed by IBM for making, reproducing, and executing quantum calculations. Qiskit ML is a segment of the Qiskit architecture and is explicitly customised for applying quantum algorithms to AI domains.

Strawberry Fields

https://strawberryfields.ai/
Developed by Xanadu, Strawberry Fields (Figure 3) is a quantum computing library focused on photonic quantum circuits. It is designed for quantum machine learning and quantum optics applications. The toolkit includes built-in functions for quantum algorithms, making it suitable for exploring quantum-enhanced machine learning methods.

QuTiP

http://qutip.org
The Quantum Toolbox in Python (QuTiP) is a versatile tool for simulating quantum systems. While it focuses on quantum dynamics and quantum optics, it can also be leveraged for QML by simulating quantum states and operations, thus supporting research in quantum-enhanced machine learning.

OpenFermion

https://quantumai.google/openfermion
This is an open source library for quantum computing in the context of quantum chemistry and fermionic systems. It integrates with other quantum computing frameworks and provides tools for simulating quantum algorithms relevant to machine learning tasks, particularly in materials science and chemistry.

ProjectQ

https://projectq.ch
ProjectQ is an open source quantum computing framework that allows users to implement quantum algorithms in Python. It provides a powerful interface for developing quantum programs and can be interfaced with existing quantum hardware and simulators. ProjectQ supports various applications, including quantum machine learning tasks, by enabling users to create and manipulate quantum circuits easily.

TensorNetwork

https://tensornetwork.org/
TensorNetwork is a library designed for the manipulation and contraction of tensor networks, which are critical for many quantum machine learning algorithms. The library is flexible and integrates well with existing machine learning frameworks, allowing researchers to explore new algorithms and models based on tensor network methods in quantum contexts.

These toolkits collectively represent a diverse set of resources available for researchers and practitioners interested in exploring the intersection of quantum computing and machine learning, facilitating innovation and experimentation in this rapidly evolving field.